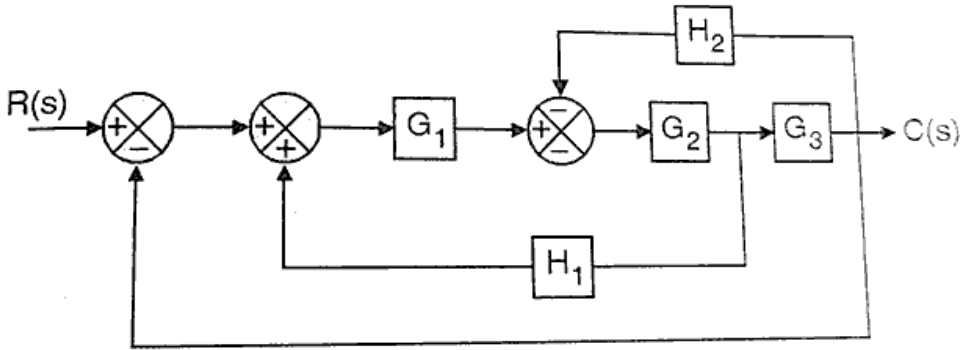
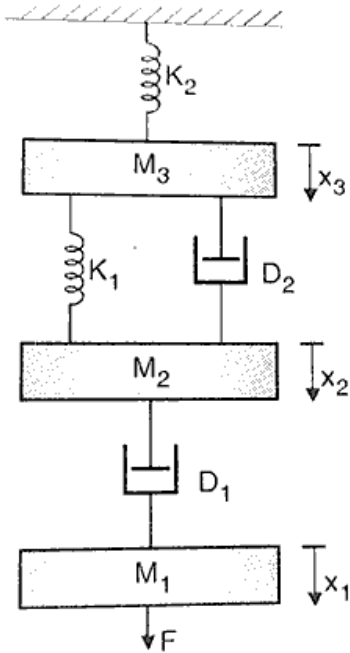
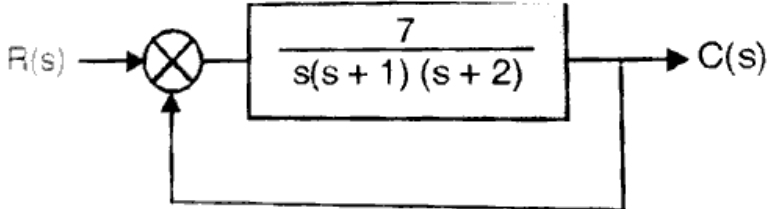


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Examination – 2023 Course: B. Tech. Branch :Electrical Semester :VI Subject Code & Name: BTEEC603 Subject: Control system Engineering Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.	CO1	12
A)	Explain signal flow graph in detail with the help of Masons gain formula.	CO1	6
B)	Using the block diagram reduction technique, find the closed loop transfer function of the system whose block diagram is given in figure. 	CO1	6
C)	Obtained the analogous electrical network for the system shown in figure. 		6

Q.2	Solve Any Two of the following.		12
A)	A unity feedback system has $G(s) = \frac{20(s+3)}{(s+1)(s+6)}$ Determine type of system, all error coefficient, error when subjected to a step of magnitude 2.	CO2	6
B)	Determine the stability using Routh's Criteria : $s^5+2s^4+3s^3+6s^2+2s+1$	CO2	6
C)	Check whether the given system is stable or not by using Hurwitz criterion. 	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Derive the transfer function of lead compensator using electrical network.	CO3	6
B)	Draw the approximate root-locus diagram for close loop system whose transfer function is given by $G(s)H(s) = \frac{k}{s(s+5)(s+10)}$	CO3	6
C)	Sketch the Bode plot showing the magnitude the decibel and phase angle in degrees as a function of log frequency for the transfer function given below. Determine the gain cross-over frequency ω_{gc} . $G(s) = \frac{10}{s(1 + 0.5s)(1 + 0.01s)}$	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Write a short note on PI controller and PID controller.	CO4	6
B)	What are the effects of proportional, integral and derivative control on the time response of the system.	CO4	6
C)	A PI controller is used to control a system. It has the following settings, $k_p = 2\%$, $k_i = 3\%$ per minutes, $m(0) = 25\%$. The error signal is found to be $4t + 2$. Find the controller output in percent after 2 minutes.	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Obtain the state model for the system with the transfer function $y(s) = \frac{3s+4}{s^2+5s+6}$	CO5	6
B)	Find the state transition matrix for, $A = \begin{bmatrix} 1 & 0 \\ -6 & -5 \end{bmatrix}$	CO5	6

C)	A linear time invariant system is characterized by the state variable model. Comment on the controllability and observability of the system. $\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$ $Y(t) = [1 \quad 2] \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$	CO5	6
	*** End ***		